HalluTruthQA: A Fine-Grained Benchmark for Hallucination Detection, Localization, and Explanation in Arabic QA

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Abstract

Large language models (LLMs) can generate fluent Arabic answers, yet factual errors remain difficult to detect, localize, explain, and verify. Existing hallucination benchmarks often provide only response-level labels, offering limited support for identifying the exact erroneous content, explaining why it is incorrect, or recovering the correct factual answer. We introduce *HalluTruthQA*, a fine-grained benchmark for hallucination evaluation in Arabic question answering. HalluTruthQA contains 2,400 expert-curated examples across four knowledge-intensive domains: Islamic knowledge, history, science, and geography. Each example includes an Arabic question, a model-generated answer, a verified reference answer, a binary hallucination label, and six candidate answers for factual verification. For hallucinated answers, the benchmark additionally provides character-level erroneous spans, human-written explanations, and macro- and micro-level hallucination types.

We evaluate four open-source LLMs—ALLaM-7B, Falcon-H1R-7B, Qwen3-32B, and SILMA—in a zero-shot setting across four tasks: hallucination detection, span-level localization, factual verification, and explanation evaluation. The results show that these tasks measure complementary capabilities, with no single model achieving the best performance across all of them. The best observed scores are 0.880 Macro-F1 for hallucination detection, 0.516 F1-Sp for span localization, 0.852 LO-Score for factual verification, and 0.644 for explanation evaluation. Our findings highlight the importance of moving beyond binary hallucination detection toward fine-grained evaluation that also localizes, verifies, categorizes, and explains factual errors. The code, dataset, prompts, and evaluation scripts are available at https://gitlab.com/nlpresearcher/HalluTruthQA.

1 Introduction

Large language models have achieved strong performance in open-domain and knowledge-intensive question answering (QA). While they generate fluent and persuasive answers, their factual reliability remains a challenge. LLMs are prone to hallucinations: statements that are unsupported, fabricated, or factually incorrect despite appearing plausible (Ji et al., 2023; Huang et al., 2025b). These errors may involve wrong entities, dates, numerical values, locations, quotations, source attributions, or explanatory claims. In QA, even a short erroneous span can make an otherwise fluent answer unreliable. Most hallucination evaluation resources rely on response-level labels indicating whether an answer is correct or hallucinated. While useful, such labels provide limited insight into the nature of the error. A response may contain the correct short answer but still include a fabricated source, an incorrect quotation, a wrong date, or an unsupported explanation. In these cases, binary evaluation fails to identify where the error occurs, why it is wrong, or whether the model can select the correct answer. This motivates a fine-grained approach to hallucination evaluation that integrates answer-level detection, span-level localization, explanation, and verification.

This problem remains underexplored in Arabic. While existing Arabic resources have made progress, they do not provide comprehensive support for knowledge-intensive QA, exact character-level span annotations, human-written explanations, and candidate-based factual verification within a single benchmark. This gap is critical, as Arabic QA often involves domains where factual accuracy depends on fine-grained details and source-sensitive verification.

Islamic knowledge, in particular, represents a sensitive setting. An answer may be superficially correct while grounding its claim in an incorrect

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Qur'anic reference, a fabricated hadith attribution, an invented quotation, or an unsupported legal or doctrinal claim. In this context, factual reliability depends not only on the final answer but also on the faithfulness of the supporting evidence. Similar challenges arise in history, science, and geography, where errors involve temporal reasoning, numerical values, spatial relations, and distinctions between closely related entities.

We introduce HALLUTRUTHQA, a fine-grained benchmark for hallucination evaluation in Arabic QA. The benchmark contains 2,400 expert-curated QA examples across four knowledge-intensive domains: Islamic knowledge, history, science, and geography. Each example pairs an Arabic question and a model-generated answer with a verified reference answer, a binary hallucination label, six candidate answers for factual verification, and, for hallucinated answers, character-level erroneous spans accompanied by human-written explanations. We evaluate four open-source LLMs in a zero-shot setting across the four tasks. Our results show that these tasks capture distinct challenges: models that perform well at detecting hallucinations do not necessarily localize erroneous spans, select the correct answer, or provide faithful explanations.

This work makes three contributions:

- We introduce HALLUTRUTHQA, a fine-grained Arabic QA hallucination benchmark with 2,400 expert-curated QA examples across four knowledge-intensive domains.
- We provide a multi-layer annotation scheme that includes response-level hallucination labels, character-level erroneous spans, human-written explanations, and six candidate answers for factual verification.
- We evaluate four open-source LLMs in a zero-shot setting across hallucination detection, span-level localization, factual verification, and explanation evaluation.

2 Related Work

Hallucination evaluation has become a central topic in large language model research. Hallucinations are commonly divided into factuality hallucinations, where generated claims contradict verifiable knowledge or introduce unsupported information, and faithfulness hallucinations, where the output is inconsistent with the input, context, instructions, or its own reasoning (Huang et al., 2025b). Existing detection approaches include fact verification, uncertainty estimation, self-consistency, textual entailment, QA-based consistency checking, and LLM-based evaluation. They differ in how precisely they identify hallucinations, from assigning a label to the whole response to detecting specific erroneous parts of the generated text.

Several benchmarks study hallucination across QA, summarization, retrieval-augmented generation, machine translation, and multilingual generation. TruthfulQA (Lin et al., 2022) uses adversarial questions designed to elicit common misconceptions, while HaluEval (Li et al., 2023) combines automatically generated and human-annotated examples. FELM (Chen et al., 2023) provides fine-grained factuality annotations across multiple domains, and RAGTruth (Niu et al., 2024) focuses on retrieval-augmented generation. Mu-SHROOM (Vazquez et al., 2025) formulates multilingual hallucination detection as span labeling, whereas HalluVerse25 (Abdaljalil et al., 2025) creates multilingual examples by injecting controlled errors into factual biographical sentences. HalOmi (Dale et al., 2023) studies hallucination and omission detection in multilingual machine translation, including Arabic, while Poly-FEVER (Zhang et al., 2025) addresses multilingual factual verification and hallucination detection across multiple languages, including Arabic. These resources differ in language coverage, task setting, grounding information, and annotation granularity, but few jointly support detection, exact localization, explanation, and factual verification. Arabic hallucination resources remain comparatively limited.

HALWASA (Mubarak et al., 2024) focuses on isolated LLM-generated sentences conditioned on predefined keywords rather than natural QA. Multilingual benchmarks provide partial Arabic coverage: HalluVerse25 includes Arabic in a controlled-error setting, while Mu-SHROOM provides span-level annotations for Modern Standard Arabic. Follow-up work has explored semantic-role decomposition and textual entailment for Mu-SHROOM (Elchafei and Abu-Elkheir, 2025); however, these multilingual settings are not specifically designed for knowledge-intensive Arabic QA. Other resources address Arabic generative QA more directly.

ISLAMICEVAL (Mubarak et al., 2025) evaluates hallucination detection and mitigation in Qur'anic and Hadith content, while AFTINA (Mohammed

et al., 2025) studies retrieval-based mitigation for Islamic fatwa generation. Their domain-specific scope, however, limits their coverage of broader Arabic knowledge. AraHalluEval (Alansari and Luqman, 2025) evaluates Arabic QA and summarization using 12 factuality and faithfulness indicators, including entity and numerical errors, contradictions, source conflicts, and fabrications. HalluScore (Alansari and Luqman, 2026a) provides a multi-domain Arabic QA benchmark covering adversarial formulations, reasoning, historical knowledge, and Arabic cultural contexts, together with verified evidence and reviewed answer explanations. Nevertheless, neither resource defines exact character-level localization as a general task for all hallucinated answers. In HalluScore, erroneous spans are recorded only for partially hallucinated responses.

Fine-grained localization is important because response-level labels cannot identify the precise nature or extent of an error. A response may contain the correct main answer while adding an incorrect entity, date, numerical value, quotation, attribution, or supporting claim. Similarly, two responses with the same hallucination label may differ substantially, with one being entirely incorrect and the other containing only a short unsupported span. This limitation is especially important in knowledge-intensive and source-sensitive domains. In legal and religious QA, for example, an answer may appear persuasive while misattributing a ruling, misquoting an authoritative text, or fabricating supporting evidence. Prior work shows that LLMs can misquote Qur'anic verses or generate unsupported evidence in Islamic answers (Bouchekif et al., 2025b). Evaluation should therefore assess not only overall correctness, but also source attribution, evidential grounding, and the exact location of erroneous content.

HalluTruthQA complements these resources by combining four evaluation dimensions for knowledge-intensive Arabic question answering: response-level hallucination detection, exact character-level span localization, human-written explanations, and candidate-based factual verification. Each model-generated answer is checked against a verified reference answer, allowing the benchmark to evaluate whether an answer is hallucinated, where the error occurs, why it is incorrect, and whether the correct factual answer can be selected. A discussion of hallucination causes and a feature-based comparison with related Arabic and multilingual resources are provided in Appendix A and Table 3.

Domain	Hallucinated	Non-hallucinated	Hallu. Spans
Islamic	341 (56.8%)	259 (43.2%)	393
History	193 (32.2%)	407 (67.8%)	199
Science	255 (42.5%)	345 (57.5%)	255
Geography	224 (37.3%)	376 (62.7%)	273
Total	**1013 (42.2%)**	**1387 (57.8%)**	**1120**

Table 1: Distribution of hallucinated and non-hallucinated responses across domains. "Hallucinated Spans" denotes the total number of distinct text segments annotated as factually incorrect or unsupported within the generated responses. A single response may contain multiple hallucinated spans.

3 Data Description

HalluTruthQA contains 2,400 Arabic QA examples across four knowledge-intensive domains: Islamic knowledge, history, science, and geography (Table 1). Each example includes an Arabic question, a model-generated answer, a verified reference answer, and a binary hallucination label. For hallucinated answers, we provide character-level erroneous spans and human-written explanations. These spans are further categorized using a two-level hallucination taxonomy. Macro-types capture failure categories, such as factual contradiction, context inconsistency, and factual fabrication, whereas micro-types identify the specific error affecting an annotated span, such as an incorrect entity, date, numerical value, citation, or source attribution. Each question is paired with six manually constructed candidate answers: one correct answer and five plausible distractors written in a style close to the generated output, so factual verification cannot rely on superficial stylistic cues. The benchmark supports hallucination detection, span-level localization, explanation evaluation, and multiple-choice factual verification.

We use Fanar-1-9B-Instruct[1] (Fanar Team et al., 2025) as the single response generator for all questions. This provides a controlled and reproducible setting, where all answers are produced by the same Arabic-centric instruction-tuned model. It also avoids additional variation that could come from using several generators with different styles, strengths, and error patterns. Since the evaluated answers are generated by FANAR-9B, we additionally report FANAR-9B as a self-detection refer-

[1] https://huggingface.co/QCRI/Fanar-1-9B-Instruct

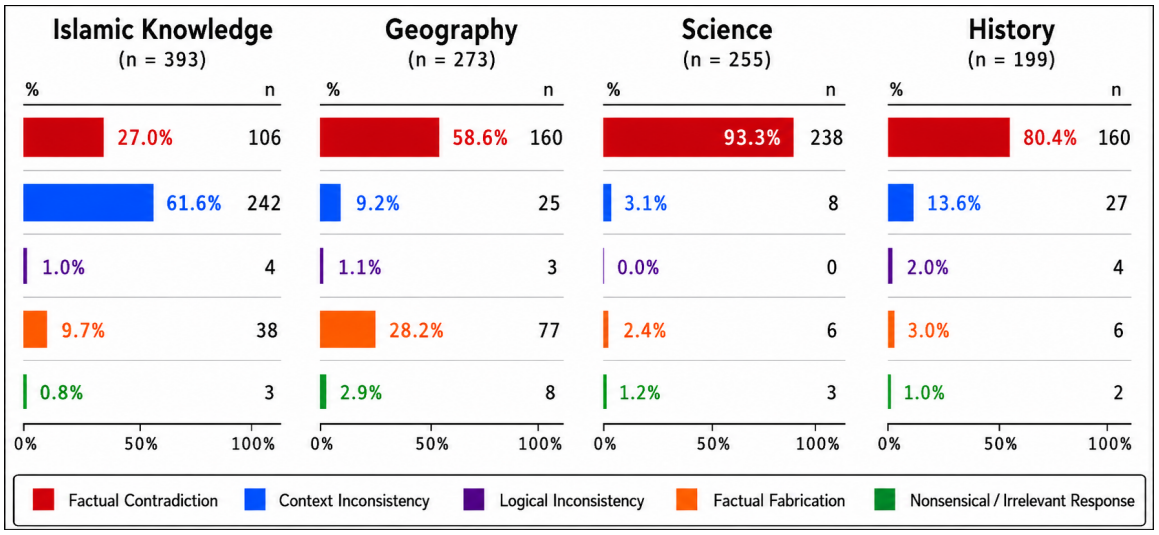


Figure 1: Distribution of hallucination macro-types across domains.

ence setting to study whether a model can detect, localize, verify, and explain hallucinations in its own outputs. These results are reported separately, rather than as evidence that FANAR-9B is superior to the other models. Answer generation used a maximum of 1024 new tokens, temperature 0.0, top-p = 1.0, batch size 4, and `bfloat16` precision on a 48 GB GPU.

3.1 Annotation Process

Annotation followed two passes: expert annotation followed by verification by trained research assistants. The four domains were handled by four experts with relevant academic backgrounds, with one expert responsible for each domain. This ensured that each subset was annotated by an expert familiar with the corresponding knowledge area and its factual verification needs. Annotators checked errors involving entities, dates, numerical values, unsupported claims, fabricated references, and incorrect attributions. For Islamic knowledge, they also verified Qur'anic references and wording, hadith attributions, and unsupported religious claims. For hallucinated responses, annotators marked character-level erroneous spans and wrote explanations. A response may contain multiple erroneous spans. Each span is annotated independently as the minimal text segment needed to isolate a single factual error. Each span is assigned one macro-type and one micro-type; multi-label assignment is therefore not used at the span level. When a response contains several distinct errors, they are annotated as separate spans.

Annotators manually created six candidate answers per question: one verified correct answer and five plausible distractors. The distractors were written in a style close to the generated response but were factually incorrect, so factual verification could not rely on superficial stylistic cues.

In the second pass, two research assistants reviewed all 2,400 examples. This pass flagged 84 examples (3.5%), which were returned to the original domain experts for adjudication. Of these, 34 were confirmed without changes, 11 questions were revised for ambiguity, and 39 examples were revised, mainly to span boundaries. Before final adjudication, expert–reviewer agreement was high across annotation dimensions, with Cohen's $\kappa = 0.93$ for binary labels, $\kappa = 0.87$ for macro-types, $\kappa = 0.81$ for micro-types, 97.9% agreement on the multiple-choice answer key, and span-level agreement of 0.89 character-F1 and 0.84 IoU. More annotation details are provided in Appendix C.

3.2 Hallucination Type Distribution

Figure 1 shows the distribution of hallucination macro-types across domains. Across all hallucinated spans, *Factual Contradiction* is the dominant category (59.3%), typically involving wrong entities, dates, locations, events, numbers, roles, or concepts. *Context Inconsistency* follows (27.0%), covering source, attribution, or evidential mismatches. *Factual Fabrication* accounts for 11.3%, while *Nonsensical / Irrelevant Response* and *Logical Inconsistency* are rare (1.4% and 1.0%). At the domain level, *Factual Contradiction* dominates in History, Science, and Geography, accounting for 80.4%, 93.3%, and 58.6% of hallucinated spans,

respectively. Islamic Knowledge differs: *Context Inconsistency* accounts for 61.6%, reflecting the need for faithful grounding in the appropriate verse, hadith, legal ruling, or source context. Geography shows the highest rate of Factual Fabrication (28.2%). Many geography questions require precise long-tail knowledge about locations and spatial relations, which may lead models to generate plausible but unsupported locations or overly specific geographic explanations. Overall, hallucinations are usually fluent and relevant, but factually incorrect, unsupported or fabricated.

These patterns highlight the need to evaluate attribution faithfulness and span-level localization beyond response-level detection. The complete hallucination taxonomy, including macro-type definitions, observed micro-types, is presented in Appendix D, with examples provided in Appendix E.

4 Experimental Setup

4.1 Evaluation Tasks

We assess model performance across four tasks:

- **Binary Hallucination Detection:** Given a question and a generated answer, the model predicts whether the response is HALLUCINATED or NON-HALLUCINATED. This task evaluates the model's ability to detect factually incorrect, fabricated, or unsupported content.
- **Span-level Localization:** For hallucinated instances, the model localizes erroneous spans at the character level. Performance is based on overlap between predicted and gold erroneous spans.
- **Explanation Evaluation:** The model generates an explanation identifying the erroneous content and justifying why it is factually incorrect. We use an LLM-as-a-judge protocol to compare generated explanations against human-written reference explanations. The judge assigns a score from 0 to 2 (0: incorrect/irrelevant; 1: partially correct; 2: complete/correct), normalized to $[0, 1]$.
- **Multiple-Choice Factual Verification:** The model selects the correct option from six candidate answers, including one correct answer and five distractors. This evaluates whether the model can identify the correct factual answer among plausible alternatives.

4.2 Models and Inference Protocol

Models. We evaluate four open-source LLMs: ALLAM-7B, FALCON-H1R-7B, QWEN3-32B, and SILMA. Since all evaluated answers are generated by FANAR-9B, we additionally report FANAR-9B as a self-detection reference setting to assess whether it can detect hallucinations in its own generated answers.

Inference Setup. All models are evaluated in a zero-shot, closed-book setting without access to external knowledge bases or web search. To ensure a fair comparison, we use identical generation parameters: temperature 0.0, top-p = 1.0, and a maximum of 1024 new tokens. We use a two-stage prompting strategy: the first prompt asks the model to perform binary hallucination detection; for cases predicted as HALLUCINATION, the second prompt asks for span localization, explanation, and factual verification.

4.3 Evaluation Metrics

We evaluate models on hallucination detection, span-level localization, factual verification, and explanation quality.

- **Hallucination Detection:** We report accuracy and F1-score for both hallucinated and non-hallucinated classes. We use Macro-F1 as the primary detection metric, as it equally weights both classes and is robust to label imbalance.
- **Span-level Localization:** We report partial-credit span-level F1 (*F1-Sp*) following (Sky et al., 2024). Although hallucinated spans are stored as character offsets, we compute *F1-Sp* after mapping predicted and gold spans to token sets. Recall is the average coverage of each gold span by the predicted spans, while precision is the average coverage of each predicted span by the gold spans. *F1-Sp* is the harmonic mean of these two values. Detection-label mismatches receive a span score of zero, while true-negative examples are excluded. This metric rewards partial span overlap without requiring exact boundary matches.
- **Multiple-Choice Factual Verification:** We report *Correct*, the number of correctly selected options among true-positive hallucination predictions, and *LO-Score* (Label–Option Score), which evaluates hallucination detection and option selection. For each example, LO-Score gives 1.0 for a correct NO-HALLUCINATION pre-

Domain	Model	Hallucination Detection				Span Localization			MCQ Selection		Explanation		
		Acc.	Hall. F1	No-Hall. F1	Macro-F1	Span P.	Span R.	F1-Sp	Correct	LO-Score	Err. Ident.	Fact. Corr.	Final Score
Islamic	QWEN3-32B	0.868	0.888	0.840	**0.864**	0.850	0.344	**0.489**	306/313	0.863	0.661	0.433	0.547
	FALCON-H1R-7B	0.863	0.888	0.826	0.857	0.769	0.312	0.444	312/324	0.853	0.659	0.494	0.576
	ALLAM-7B	0.862	0.876	0.844	0.860	0.839	0.248	0.383	247/293	0.823	0.668	0.530	**0.599**
	SILMA	0.737	0.713	0.757	0.735	0.802	0.102	0.181	81/196	0.641	0.243	0.119	0.181
	FANAR-9B	0.908	0.924	0.885	0.904	0.823	0.319	0.460	303/333	0.883	0.627	0.354	0.490
Geography	QWEN3-32B	0.895	0.870	0.912	**0.891**	0.635	0.409	**0.498**	207/211	0.892	0.594	0.584	**0.589**
	FALCON-H1R-7B	0.853	0.825	0.874	0.849	0.463	0.410	0.435	202/208	0.848	0.593	0.554	0.574
	ALLAM-7B	0.877	0.832	0.903	0.867	0.500	0.254	0.337	112/183	0.818	0.522	0.520	0.521
	SILMA	0.820	0.697	0.872	0.784	0.524	0.166	0.252	76/124	0.780	0.341	0.321	0.331
	FANAR-9B	0.877	0.835	0.902	0.868	0.510	0.254	0.340	175/187	0.867	0.517	0.499	0.508
Science	ALLAM-7B	0.915	0.908	0.921	0.915	0.766	0.344	**0.475**	243/253	0.907	0.761	0.723	**0.742**
	QWEN3-32B	0.778	0.793	0.762	0.777	0.570	0.388	0.462	254/254	0.778	0.688	0.485	0.587
	FALCON-H1R-7B	0.885	0.880	0.889	0.885	0.663	0.344	0.453	254/254	0.885	0.675	0.510	0.592
	SILMA	0.955	0.947	0.961	**0.954**	0.878	0.207	0.335	150/241	0.879	0.503	0.390	0.446
	FANAR-9B	0.965	0.960	0.969	0.965	0.849	0.408	0.551	253/255	0.963	0.685	0.466	0.575
History	QWEN3-32B	0.855	0.812	0.881	0.847	0.545	0.660	**0.597**	175/188	0.844	0.689	0.611	0.650
	ALLAM-7B	0.873	0.827	0.899	0.863	0.557	0.594	0.575	120/182	0.821	0.711	0.694	**0.702**
	FALCON-H1R-7B	0.836	0.794	0.864	0.829	0.513	0.634	0.567	172/189	0.822	0.687	0.623	0.655
	SILMA	0.881	0.819	0.912	**0.865**	0.484	0.362	0.415	97/161	0.828	0.263	0.246	0.255
	FANAR-9B	0.902	0.863	0.923	0.893	0.634	0.696	0.664	166/186	0.885	0.683	0.521	0.602
Global	QWEN3-32B	0.849	0.842	0.855	0.849	0.656	0.426	**0.516**	942/966	0.844	0.659	0.514	0.587
	FALCON-H1R-7B	0.859	0.853	0.866	0.859	0.618	0.400	0.486	940/975	0.852	0.654	0.536	0.595
	ALLAM-7B	0.882	0.865	0.894	**0.880**	0.688	0.333	0.449	722/911	0.842	0.673	0.614	**0.644**
	SILMA	0.848	0.799	0.878	0.838	0.699	0.188	0.296	404/722	0.782	0.351	0.272	0.312
	FANAR-9B	0.913	0.902	0.922	0.912	0.727	0.390	0.508	897/961	0.900	0.632	0.444	0.538

Table 2: Bold indicates the best performance for each metric within each domain and overall among the independent evaluation models; FANAR-9B is reported separately as a self-detection reference.

diction, 0.5 for a correct HALLUCINATION prediction with an incorrect option, 1.0 for a correct HALLUCINATION prediction with the correct option, and 0.0 for incorrect label prediction.

- **Explanation Evaluation.** We evaluate explanation quality using GPT-5.5 as an LLM judge. The judge is given the question, the model answer, the gold erroneous span, the human-written reference explanation, and the model explanation. It assigns a score reflecting how well the model justifies why the response is hallucinated. Explanation quality is computed only when both the gold label and the model prediction are hallucinated. The prompt is provided in Appendix F. To check the reliability of the scores, two annotators reviewed a random sample of 25 explanations per domain. This review did not reveal systematic disagreement between the judge scores and the human-written explanations.

Because span localization, factual verification, and explanation generation are performed only after a model predicts HALLUCINATION, these scores should be interpreted as pipeline metrics. Errors in binary hallucination detection can therefore propagate to the downstream tasks.

4.4 Results Analysis

Table 2 compares model performance across hallucination detection, span localization, multiple-choice factual verification, and explanation quality. Since all evaluated answers were generated by FANAR-9B, its self-detection result may partly reflect cues from its own writing style and the way it expresses uncertainty in generated answers. We therefore report FANAR-9B separately as a reference setting, rather than as direct evidence of general superiority over the other models.

Among the independent models, ALLAM-7B shows the strongest detection performance, achieving the highest global Macro-F1 score (0.880) with balanced performance across the two classes. FALCON-H1R-7B and QWEN3-32B obtain slightly lower Macro-F1 scores, with 0.859 and 0.849, respectively. SILMA shows a different behavior: its lower hallucination F1 compared to its non-hallucination F1 indicates that it is more conservative and misses more hallucinated responses.

Span-level localization shows that detecting a hallucinated answer is not sufficient: models must also identify the erroneous portion of the response. Under the partial-credit F1-Sp metric, QWEN3-32B achieves the best global localization performance, with an F1-Sp of 0.516, a precision of 0.656, and the highest recall of 0.426. FANAR-

9B follows closely with an F1-Sp of 0.508 and obtains the highest span precision (0.727), although its lower recall (0.390) indicates that it localizes fewer gold erroneous spans. FALCON-H1R-7B ranks third with an F1-Sp of 0.486, followed by ALLAM-7B with 0.449. These results reveal a precision–recall trade-off: although FANAR-9B and ALLAM-7B produce precise spans, they leave a substantial proportion of the annotated erroneous content uncovered. SILMA remains the weakest localization model, with an F1-Sp of 0.296 and particularly low recall (0.188), consistent with its conservative hallucination detection behavior.

The low F1-Sp scores show that span localization is substantially harder than binary hallucination detection. The difficulty is not only deciding whether an answer is hallucinated, but identifying the exact part of the Arabic response that makes it wrong. Many errors involve short factual units, such as dates, numbers, entities, verse references, or source attributions, where small boundary differences can reduce overlap. In other cases, the correct short answer is followed by hallucinated support, so models may select only the answer entity or miss the false supporting claim. Arabic also introduces boundary variation because clitics, attached particles, and surface forms can make character-level spans harder to align exactly. As a result, models may detect the hallucination correctly but produce spans that are too short, too broad, or shifted from the erroneous span.

For MCQs, FALCON-H1R-7B performs best among the independent models, with a global LO-Score of 0.852, closely followed by QWEN3-32B (0.844) and ALLAM-7B (0.842). FALCON-H1R-7B is effective at selecting the correct candidate answer once an error is identified, selecting the wrong option in only 3.6% of true-positive cases.

In terms of explanation quality, ALLAM-7B outperforms all evaluated systems, including FANAR-9B. It achieves the highest Error Identification (0.673) and Factual Correction (0.614) scores, leading to the best Final Score of 0.644. This highlights a difference between detection and explanation: while FANAR-9B excels at hallucination detection and candidate-option selection and remains competitive in span localization, ALLAM-7B provides stronger natural-language explanations with clearer factual corrections.

Domain-Specific Insights. The domain-level results show that hallucination evaluation presents different challenges across knowledge areas. In the Islamic Knowledge domain, factual correction and evidence grounding are difficult for all models, as errors often involve scriptural wording, hadith attribution, transmission-related information, or source-specific religious context. Although FANAR-9B and QWEN3-32B achieve strong detection performance, their factual-correction scores remain lower than in other domains. This suggests that detecting a hallucination does not necessarily mean that a model can recover or justify the correct religious evidence. For span localization in Islamic Knowledge, QWEN3-32B obtains the highest partial-credit F1-Sp score (0.489), followed by FANAR-9B (0.460), FALCON-H1R-7B (0.444), and ALLAM-7B (0.383). The relatively high span precision of QWEN3-32B (0.850), ALLAM-7B (0.839), and FANAR-9B (0.823), together with much lower recall, indicates that models often identify accurate erroneous spans when they make a prediction, but miss a large portion of the full annotated error. Overall, models can localize some explicit fabricated references or incorrect religious claims, but they struggle to cover all erroneous content and to produce factual corrections.

In Geography, QWEN3-32B achieves the strongest overall performance, leading in Macro-F1, F1-Sp, LO-Score, and explanation quality. Its F1-Sp score reaches 0.498, outperforming FALCON-H1R-7B (0.435), FANAR-9B (0.340), and ALLAM-7B (0.337). This suggests that QWEN3-32B is more effective at localizing errors involving geographic locations and spatial claims.

In Science, detection performance is generally high across models. Including the self-detection reference setting, FANAR-9B achieves the highest Macro-F1 score (0.965) and the highest F1-Sp score (0.551). Among the independent evaluation models, SILMA achieves the strongest detection performance with a Macro-F1 of 0.954, while ALLAM-7B obtains the best span localization score with an F1-Sp of 0.475 and the best explanation Final Score of 0.742. These results suggest that scientific hallucinations often involve erroneous values, entities, formulas, or causal claims, which can be localized more easily than implicit or context-dependent errors. However, substantial portions of the annotated erroneous spans remain uncovered.

In History, FANAR-9B achieves the best localization performance in the self-detection reference

setting, with an F1-Sp score of 0.664. QWEN3-32B obtains the highest F1-Sp score (0.597), followed by ALLAM-7B (0.575) and FALCON-H1R-7B (0.567). The higher recall of FANAR-9B (0.696) and QWEN3-32B (0.660) indicates that these models capture a larger share of historically erroneous content. At the same time, ALLAM-7B provides the strongest explanation quality, with the highest Final Score of 0.702.

Error Analysis. To better understand the types of errors, we analyze the 790 unique instances where at least one model makes a wrong prediction. Since the same instance can be misclassified by more than one model, these instances produce 1,556 model-level errors in total. We group the most frequent errors into four main families:

1. *Temporal, Numeric, and Unit Normalization (39.0%):* These errors occur when models fail to recognize that two different formats express the same fact. For example, a model may mark an answer as hallucinated because it says "40 AH", while the reference answer says "661 CE". Although the formats are different, they refer to the same historical period. This shows that models still struggle with converting dates, numbers, units, and other equivalent factual expressions (see Example Non-Hallucinated, Example 2, Example 3, and Example 17).
2. *Source Attribution and Evidence Verification (22.2%):* These errors are especially common in the Islamic domain. They occur when the model gives the correct main answer but supports it with an incorrect source, such as a fabricated verse reference, a wrong citation, or an inaccurate attribution. In these cases, models often focus on the fact that the short answer is correct, but they fail to verify whether the supporting evidence is also correct (see Example 11 and Example 15).
3. *Correct Short Answer with Hallucinated Support (17.1%):* These errors occur when the model gives the correct short answer but adds a false or unsupported explanation. For example, the named entity or final answer may match the reference answer, but the justification contains an incorrect claim. Detection models often miss these cases because they rely too much on the overlap with the correct answer and ignore the hallucinated information in the explanation (see Example 9 and Example 10).
4. *Fully Incorrect Answer (11.3%):* These errors occur when the model's main answer is factually incorrect. In some cases, the generated answer may be entirely different from, or directly contradict, the correct answer. The error affects the core answer rather than only the explanation or supporting evidence (see Example 7, Example 8, and Example 11).

Together, these four families cover 89.6% of the model-level errors. The remaining 10.4% correspond to mixed or less frequent cases that combine several patterns, such as partial factual mismatches, weak evidence checking, over-specific additions, or imprecise span localization. We do not treat these cases as a separate family because they do not form a single recurring error pattern.

Common Failure Cases. A total of 39 examples triggered an identical failure across all five evaluated systems, including FANAR-9B in the self-detection reference setting. The vast majority of these shared errors (79.5%) are false positives, where a valid response was uniformly misclassified as hallucinated. Only 20.5% correspond to missed true hallucinations. These shared failures are concentrated in History (18 cases), followed by Geography (10), Islamic (8), and Science (3).

These findings emphasize that resolving hallucinations in advanced Arabic models requires moving beyond factual recall; models must recognize semantic paraphrases, validate fine-grained source attributions, handle temporal and numerical conversions, isolate correct short answers from faulty supporting explanations, and localize the exact erroneous spans. Additional annotation and evaluation details are provided in Appendix C.

5 Conclusion

We introduced HALLUTRUTHQA, a fine-grained, expert-curated benchmark for evaluating hallucinations in Arabic QA. Spanning Islamic knowledge, history, science, and geography, HALLUTRUTHQA provides response-level hallucination labels, character-level erroneous spans, human-written explanations, and multiple-choice factual verification candidates. This multidimensional design shifts hallucination evaluation from binary correctness toward a granular understanding of where and why models fail, and whether they can recover the correct factual information. Our evaluation of four open-source instruction-tuned LLMs, together with the FANAR-9B self-detection reference setting, shows that hallucina-

tion detection, span localization, factual verification, and explanation quality remain distinct, non-trivial challenges. A key finding is that strong detection performance does not necessarily imply high-quality explanations or accurate span localization: FANAR-9B performs strongly in the self-detection reference setting, QWEN3-32B achieves the best global span localization, and ALLAM-7B provides the most reliable factual explanations. Moreover, our results reveal domain-specific patterns: Islamic Knowledge is affected by evidence-grounding and source-attribution errors, whereas history, science, and geography are mainly affected by factual contradictions and fabricated details. These findings underscore the need for Arabic hallucination benchmarks that move beyond binary detection to assess attribution faithfulness, temporal consistency, and reasoning errors. HALLUTRUTHQA provides a step toward more reliable Arabic QA systems. Future work will expand the benchmark to additional domains, broader model coverage, and training settings aimed at improving factual reliability in Arabic language models.

Limitations

This work has several limitations.

- First, HALLUTRUTHQA focuses on four knowledge-intensive domains: Islamic knowledge, history, science, and geography. Although these domains cover diverse factual phenomena, the benchmark does not capture all types of Arabic QA, such as legal, medical, financial, or conversational settings.
- Second, the model-generated answers in the dataset are produced using a single LLM, FANAR-9B. This design enables controlled annotation and analysis, but it may limit the diversity of hallucination patterns compared to datasets built from multiple generation models. Future work can extend the benchmark with answers generated by a range of Arabic and multilingual LLMs.
- Third, span-level hallucination annotation is challenging. In some cases, hallucinated content may involve implicit reasoning errors, unsupported claims, or source attribution errors that are difficult to isolate with exact character boundaries. To reduce this issue, we rely on expert validation and use overlap-based span evaluation, but boundary variation remains possible.
- Fourth, explanation quality is evaluated using an LLM-as-a-judge protocol, with additional human validation on a sample of judged explanations. While this helps make the evaluation scalable, automatic judging may still introduce bias or inconsistency. More extensive human evaluation would further strengthen the reliability of explanation assessment.
- Finally, we deliberately focus on open models and exclude proprietary API-based systems such as Gemini or Claude from the main evaluation. This choice is motivated by fairness and reproducibility. Our benchmark is designed as a closed-book evaluation, where all models are tested under the same inference conditions and without access to external search. In contrast, API-based systems may rely on changing backend versions, retrieval components, web search, or system-level grounding that are not always fully controllable by researchers. Including such systems could make it harder to ensure that the evaluation measures the model itself rather than external information access. This design also supports the analysis and improvement of open Arabic language models, which can be inspected, reproduced, and adapted by the research community.

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A Hallucination in Large Language Models and Related Resources

Definition. LLMs can generate fluent and convincing answers that are not necessarily factually

correct. Hallucination generally refers to generated content that is inaccurate, fabricated, misleading, unsupported by reliable evidence, or inconsistent with the provided input or context (Ji et al., 2023; Huang et al., 2025a). Hallucinations are commonly divided into two broad categories. A *factual hallucination* occurs when a generated statement contradicts verifiable real-world knowledge, invents an entity or event, or incorrectly attributes a fact to a person, place, time, or source. A *faithfulness hallucination* occurs when the generated content is not supported by the input or the provided context, even when the statement may be factually correct in isolation.

Causes of hallucination. Hallucination should not be attributed to a single cause. It can emerge at different stages of the LLM development and generation pipeline (Huang et al., 2025a; Alansari and Luqman, 2026b). One important cause is incomplete knowledge coverage: the information required to answer a question may be absent from, rare in, or insufficiently represented in the model's training corpus. As a result, the model may rely on related language patterns and generate a plausible but unsupported answer. Hallucinations may also arise from incorrect, outdated, noisy, or conflicting information in the training data, which can lead the model to learn unreliable associations.

This issue is particularly relevant for Arabic, where high-quality, diverse, and domain-specific data remain limited, especially in specialized areas such as Islamic knowledge and legal reasoning. Several Arabic-language resources have been introduced to evaluate LLMs on Islamic knowledge, legal reasoning, and general culture (Alwajih et al., 2025; Bouchekif et al., 2025a, 2026b,a; Gaben et al., 2026; Eltanbouly et al., 2026). However, many of these resources rely mainly on multiple-choice questions, where models are evaluated by selecting the correct option. This evaluation setting does not capture how models formulate complete answers. Without evaluating full generated responses, it is difficult to assess model behavior in depth, particularly whether the model introduces hallucinated claims, unsupported explanations, fabricated evidence, or incorrect source attributions.

The next-token prediction objective may further contribute to hallucination because it encourages the model to generate linguistically likely continuations rather than explicitly verify the factual correctness of each statement. Fine-tuning biases, alignment data, prompt ambiguity, insufficient contextual grounding, and decoding decisions may also increase the likelihood of hallucination. Therefore, a model may hallucinate not only because it has not encountered the required information, but also because it fails to retrieve, distinguish, or correctly use knowledge that was present during training.

Related Arabic and multilingual resources. Several recent studies have investigated hallucination in Arabic and multilingual LLMs from complementary perspectives. AraHalluEval introduces a multidimensional manual evaluation framework for Arabic generative QA and summarization, using fine-grained factuality and faithfulness indicators (Alansari and Luqman, 2025). HalluVerse-M^3 provides multilingual and multitask examples constructed through controlled editing and human validation (Abdaljalil et al., 2026). HalluScore focuses on hallucination-prone Arabic questions across multiple domains, cultural settings, and reasoning requirements, and provides fine-grained labels and explanations (Alansari and Luqman, 2026a). In the Islamic domain, Aftina studies hallucination mitigation in fatwa generation using retrieval-augmented generation and re-ranking (Mohammed et al., 2025), while IslamicEval evaluates the identification, validation, and correction of Qur'anic and Hadith quotations (Mubarak et al., 2025).
These resources differ in their languages, domains, annotation procedures, and levels of supervision. Importantly, AraHalluEval and HalluScore already go beyond binary response-level labels by providing fine-grained annotations, and HalluScore additionally provides explanations. However, neither resource provides exact character-level localization of the hallucinated content within generated responses. Span annotations are available in IslamicEval, but they identify complete intended Qur'anic and Hadith quotation spans rather than the specific erroneous segments within the overall response. In contrast, HalluTruthQA directly annotates the exact character-level spans responsible for factual errors and associates them with fine-grained hallucination types, human-written explanations, and verified factual answers. Table 3 summarizes these differences.

Overall, the existing resources address comple-

Table 3: Feature-based comparison of HALLUTRUTHQA with recent Arabic and multilingual hallucination resources. "Partial" indicates that the feature is available only for a subset of the data or is used only for system-level evaluation.

Resource	Language(s)	Domain / Task	Outputs	Manual annotation	Span annotation	Fine-grained labels	Explanations	Verified answer / Correction
ARAHALLUEVAL (Alansari and Luqman, 2025)	Arabic	Generative QA and abstractive summarization	LLM outputs	Yes; two annotators and expert adjudication	No	12 factuality and faithfulness indicators	No error-specific explanations	Reference answer for QA or source document for summarization
HALLUVERSE-M^3 (Abdaljalil et al., 2026)	English, Arabic, Hindi, and Turkish	QA and dialogue summarization	Controlled LLM-edited outputs	Human validation	No	Entity-, relation-, and sentence-level hallucination types	No	Original ground-truth output
HALLUSCORE (Alansari and Luqman, 2026a)	Arabic	Multi-domain generative QA	LLM outputs	Yes; human annotation of model responses	Partial; hallucinated text is recorded only for partial-hallucination cases, not as a general character-offset localization task	Factual, faithfulness, and partial hallucination labels, together with multi-label question metadata	Ground-truth answer explanations generated by an LLM and manually reviewed	Verified ground-truth answer and supporting source link
AFTINA (Mohammed et al., 2025)	Arabic	Islamic fatwa QA	LLM outputs + RAG	Partial; expert evaluation of 210 generated outputs	No	No corpus-level hallucination taxonomy	No	Dar Al-Ifta QA corpus and retrieved supporting content
ISLAMICEVAL (Mubarak et al., 2025)	Arabic	Qur'an and Hadith	LLM outputs	Yes; Islamic-studies experts	Yes; intended Qur'anic and Hadith quotation spans	Correct Qur'an, Incorrect Qur'an, Correct Hadith, and Incorrect Hadith	No	Canonical correction for incorrect quotations, or an indication that no valid correction exists
HALLUTRUTHQA Arabic	**Islamic knowledge, history, science, and geography**	**LLM outputs**	**Yes; domain-expert annotation followed by independent verification**	**Yes; exact character-level erroneous spans**	**Macro- and micro-level hallucination types**	**Human-written explanations of the localized errors**	**Verified reference answer and six candidate answers**	

mentary aspects of hallucination evaluation. ARAHALLUEVAL provides broad multidimensional evaluation across two Arabic generation tasks, while HALLUVERSE-M^3 supports controlled comparisons across languages and tasks. HALLUSCORE emphasizes hallucination-prone Arabic questions and provides verified evidence, answer explanations, and human response-level annotations. AFTINA primarily evaluates hallucination mitigation through retrieval and re-ranking, whereas ISLAMICEVAL specializes in the accurate identification and correction of Qur'anic and Hadith quotations. In contrast, HALLUTRUTHQA jointly supports response-level detection, exact erroneous-span localization, macro- and micro-level error classification, human-written explanation, and factual-answer verification in knowledge-intensive Arabic QA.

A.1 Question Construction and Verification

All questions were manually prepared by domain experts based on their knowledge and experience. For each question, the corresponding expert verified the reference answer using reliable sources. Experts were instructed to formulate clear, knowledge-intensive questions while maintaining diversity in topics and difficulty levels. Table 4 provides representative examples of the sources consulted for each domain; these examples are not intended to constitute an exhaustive list. For Islamic Knowledge, experts additionally checked Qur'anic wording and references, hadith attributions, and the factual support of religious claims. During verification, questions and reference answers were reviewed for accuracy, clarity, and ambiguity before inclusion in the benchmark.

After the initial annotation, two trained research assistants reviewed the questions, reference answers, hallucination labels, localized spans, explanations, and answer options. Unclear or disputed cases were returned to the original domain expert for final verification. The annotation process was based on meaning rather than exact string matching. Therefore, semantically equivalent answers were accepted when they expressed the same fact. This also applied to different date formats. For example, corresponding Hijri and Gregorian dates were accepted when the experts judged them to refer to the same historical event. In the example provided in Appendix A, "40 AH" and "661 CE" were treated as equivalent. Since both the questions and hallucination annotations were manually reviewed, differences between Hijri and Gregorian date formats were not automatically labeled as hallucinations.

B Data Format

Each instance is stored in a structured JSON format. The format includes the question, generated response, reference answer, response-level hallucination label, span-level annotations, explanations, multiple-choice candidate answers, and the correct option.

Domain	Representative Sources
Islamic Knowledge	IslamWeb; IslamQA; سؤال وجواب في القرآن الكريم 1213 (*1,213 Questions and Answers on the Holy Qur'an*) by Ahmad bin Ali Abu Islam; سؤال وجواب في العقيدة الإسلامية 200 (*200 Questions and Answers on Islamic Creed*) by Hafiz bin Ahmad Al-Hakami; السيرة النبوية في سؤال وجواب (*The Prophetic Biography in Questions and Answers*) by Abu Abdullah Muhammad Ali Samak; أكثر من 1500 سؤال وجواب في القرآن الكريم (*More Than 1,500 Questions and Answers on the Holy Qur'an*) by Muhsin Hussein Al-Ghamdi; and الموسوعة القرآنية (*The Qur'anic Encyclopedia*).
History	أطلس تاريخ الإسلام (*Atlas of Islamic History*); المفصل في تاريخ العرب قبل الإسلام (*The Detailed History of the Arabs Before Islam*).
Geography	أطلس الوطن العربي (*Atlas of the Arab World*) by Yahya Nabhan; جغرافية العالم الإسلامي (*Geography of the Islamic World*) by Mahmoud Taha Abu Al-Ala; and الموسوعة العربية العالمية (*The Global Arabic Encyclopedia*).
Science	موسوعة العلوم والتقانات (*Encyclopedia of Science and Technology*); الموسوعة العلمية الميسرة (*The Concise Scientific Encyclopedia*), edited and reviewed by Ahmad Shafiq Al-Khatib; and الموسوعة العربية العالمية (*The Global Arabic Encyclopedia*).

Table 4: Representative sources consulted by domain experts during question construction and reference-answer verification.

Hallucinated Answer


```
{
  "id": "islamicHallucination_8",
  "question":
                                    "ما البلد الذي ذُكر اسمه في القرآن الكريم؟",
  "generated_answer":
"بلد واحد فقط ذُكر اسمه في القرآن الكريم وهو مصر، وذلك في سورة يوسف الآية 47:
وَقَالَ الْمَلِكُ ائْتُونِي بِهِ فَلَمَّا جَاءَهُ الرَّسُولُ قَالَ ارْجِعْ إِلَى رَبِّكَ فَسْأَلْهُ مَا لِذِي النِّسْوَةِ إِنَّ
قَدْ ظَلَمْتُ إِنَّ",
  "gold_answer":
"البلد الذي ذُكر اسمه صراحة في القرآن الكريم هو مصر، ومن ذلك قوله تعالى: «وَقَالَ
الَّذِي اشْتَرَاهُ مِن مِّصْرَ لِامْرَأَتِهِ أَكْرِمِي مَثْوَاهُ»، وقوله تعالى: «ادْخُلُوا مِصْرَ إِن شَاءَ اللَّهُ
آمِنِينَ",
  "generator_model": "QCRI/Fanar-1-9B-Instruct",
  "label": "hallucination",
  "hallucinations": [
    {
      "span_start": 67,
      "span_end": 75,
      "hallucinated_span":
                                                "الآية 47",
      "explanation":
        "الإحالة إلى الآية غير صحيحة؛ اسم مصر لم يرد في يوسف: 47؛ وإنما ورد في
                                مواضع أخرى من السورة، منها يوسف: 21 ويوسف: 99."
    },
    {
      "span_start": 77,
      "span_end": 226,
      "hallucinated_span":
        "وَقَالَ الْمَلِكُ ائْتُونِي بِهِ فَلَمَّا جَاءَهُ الرَّسُولُ قَالَ ارْجِعْ إِلَى رَبِّكَ فَسْأَلْهُ مَا لِذِي
        النِّسْوَةِ إِنَّ قَدْ ظَلَمْتُ إِنَّ",
      "explanation":
        "هذا نص مخترع منسوب إلى القرآن، وفيه ألفاظ غير قرآنية مثل النسوة؛ والآية
        الصحيحة في يوسف: 47 تتحدث عن الزراعة والسنين: قال تزرعون سبع سنين
                                                دأبا."
    }
  ],
  "options": {
    "A":
      "- البلد الذي ذُكر اسمه صراحة في القرآن الكريم هو مصر، وقد ورد اسمها في سياق
      قصة يوسف عليه السلام.",
    "B":
      "- البلد الذي ذُكر اسمه صراحة في القرآن الكريم هو فلسطين، وقد ورد ذكرها في
      سياق الأرض المقدسة.",
    "C":
      "- البلد الذي ذُكر اسمه صراحة في القرآن الكريم هو العراق، وقد ارتبط بعدد من
      قصص الأنبياء والأمم السابقة.",
    "D":
      "- البلد الذي ذُكر اسمه صراحة في القرآن الكريم هو الأردن، وقد ارتبط بأحداث
      تاريخية وقعت في بلاد الشام.",
    "E":
      "- البلد الذي ذُكر اسمه صراحة في القرآن الكريم هو اليمن، وقد ارتبط بقصة سبأ
      وملكة سبأ.",
    "F":
      "- البلد الذي ذُكر اسمه صراحة في القرآن الكريم هو السعودية، وقد ارتبط بذكر مكة
      المكرمة والبيت الحرام."
  },
  "answer": "A"
}
```


For non-hallucinated responses, the span list is empty. This unified format allows the same dataset to be used for response-level hallucination detection, explainable hallucination analysis, span-level hallucination localization, multiple-choice factual verification, and training models to select or generate factually reliable answers.

Non-Hallucinated Answer


```
{
  "id": "historicHallucination_474",
  "question":
                                    "في أي عام انتهى عصر الخلفاء الراشدين؟",
  "generated_answer":
                                "انتهى عصر الخلفاء الراشدين في العام 40 هـ.",
  "gold_answer":
                                                "661 م",
  "generator_model": "QCRI/Fanar-1-9B-Instruct",
  "label": "no_hallucination",

}
```



Figure 2: Overview of the HALLUTRUTHQA construction pipeline.

C Annotation Process and Reliability

The annotation process involved two stages. First, four domain experts annotated the dataset, with each expert responsible for one domain: Islamic knowledge, history, science, or geography. Each domain expert had a relevant academic background in the corresponding knowledge area.

The experts manually annotated the binary hallucination label, hallucination macro-type, hallucination micro-type, erroneous spans, explanations, and reference answers. They also manually constructed and validated the multiple-choice candidate options and the corresponding answer key. Annotators were instructed to ensure that the candidate options followed the same style, wording, and level of specificity as the model-generated answer, so that the correct option could not be identified from stylistic cues.

Second, two trained research assistants independently verified the annotations produced by the four domain experts. The verification pass checked the binary hallucination label, hallucination categories, span boundaries, explanations, and the correctness of the multiple-choice answer key. Since the four domain experts annotated disjoint domain-specific subsets, agreement was not computed among the four experts on shared items. Instead, we report expert–reviewer agreement between the initial domain-expert annotations and the independent verification pass.

Disagreements or flagged cases were resolved through adjudication by the original domain expert after the second verification pass. Agreement was computed before final expert adjudication. We used Cohen's κ for categorical annotations and character-level F1 and intersection-over-union (IoU) for span-level annotations. Disagreements and flagged cases were subsequently resolved through adjudication by the original domain expert after the second verification pass.

Annotation dimension	Metric	Score
Binary hallucination label	Cohen's κ	0.93
Macro hallucination type	Cohen's κ	0.87
Micro hallucination type	Cohen's κ	0.81
Multiple-choice answer key	Agreement	0.979
Span localization	Character-F1	0.89
Span overlap	IoU	0.84

Table 5: Expert–reviewer agreement for HalluTruthQA before final adjudication. Cohen's κ is reported for categorical annotations, agreement is reported for the multiple-choice answer key, and character-level F1 and IoU are used for span-level annotations.

The agreement results indicate that the annotation process was reliable across all dimensions. Binary hallucination labeling achieved the highest agreement ($\kappa = 0.93$), showing that the domain-expert annotations and the independent verification pass were highly consistent in deciding whether a generated answer contained hallucinated content. Agreement remained strong for macro-level hallucination categories ($\kappa = 0.87$), suggesting that the main hallucination types were clearly defined and consistently applied.

Agreement was slightly lower for micro-level categories ($\kappa = 0.81$), which is expected given the difficulty of distinguishing closely related fine-grained error types, such as unsupported claims, wrong attribution, citation mismatch, and over-specific additions. Nevertheless, this score still indicates strong agreement for fine-grained hallucination classification.

The multiple-choice answer key achieved 97.9% agreement, indicating that the correct option was generally unambiguous. For span-level annotation, the character-level F1 of 0.89 and IoU of 0.84 show that the reviewers largely identified the same erroneous content as the original expert annotations. Remaining differences were mainly due to span-boundary variation, such as whether only the minimal erroneous entity, date, or reference was marked, or whether surrounding explanatory context was also included.

Overall, these results confirm that the annotations are robust. They also show that response-level hallucination detection is the most stable annotation task, whereas micro-type classification and exact span localization are more challenging but still highly reliable.

D Hallucination Taxonomy

Figure 3 presents the hierarchical taxonomy of hallucination types used in HALLUTRUTHQA. The taxonomy was designed to describe the erroneous spans observed during annotation. It contains five macro-types: *Factual Contradiction*, *Context Inconsistency*, *Logical Inconsistency*, *Factual Fabrication*, and *Nonsensical / Irrelevant Response*. Each macro-type is further divided into fine-grained micro-types that describe the specific form of the error.

Purpose of the taxonomy The goal of the taxonomy is to move beyond a binary distinction between hallucinated and non-hallucinated answers. In Arabic QA, an answer can be fluent and relevant but still contain a small factual error, a wrong source attribution, an unsupported explanation, or a fabricated reference. For this reason, each hallucinated answer in HALLUTRUTHQA is annotated not only with a response-level label, but also with character-level erroneous spans, explanations, and hallucination types.

Macro and micro types Table 6 defines the macro-types and their corresponding micro-types. The macro-types capture the main ways in which a model answer can fail, while the micro-types provide a more precise description of the erroneous span. For example, a factual contradiction may involve an entity, date, numeric, or definition error, whereas a context inconsistency may involve a citation mismatch or a wrong source attribution.

E Hallucination Examples

To make the taxonomy easier to interpret, we provide examples for the hallucination micro-types observed in HALLUTRUTHQA. Each example is labeled with its hallucination type and includes the question, the model prediction, the verified answer, and a short explanation of the error.

These examples illustrate how different hallucination types appear in Arabic QA. Some cases involve direct factual contradictions, such as wrong entities, dates, numbers, or locations. Other cases involve context inconsistency, where the answer remains related to the question but the supporting evidence, citation, or source attribution is incorrect. We also include examples of logical inconsistency, factual fabrication, and nonsensical or irrelevant responses.

1 Factual Contradiction – Entity Error

Question

من هو الصحابي الذي دعا له النبي صلى الله عليه وسلم بقوله: اللهم فقهه في الدين؟

Who was the Companion for whom the Prophet, peace be upon him, supplicated: "O Allah, grant him understanding of the religion"?

Prediction

معاذ بن جبل رضي الله عنه

Muadh ibn Jabal, may Allah be pleased with him.

Gold answer

Abdullah Ibn Abbas الاسم هو عبد الله بن عباس

Comment

The model predicts the wrong Companion instead of Abdullah Ibn Abbas.

2 Factual Contradiction – Date / Temporal Error

Question

في أي عام تم توحيد ألمانيا؟

In which year was Germany unified?

Prediction

تم توحيد ألمانيا في العام 1871.

Germany was unified in the year 1871.

Gold answer

1990

Comment

Although the German Empire was founded in 1871, the term "German unification" in contemporary contexts refers to the reunification of East and West Germany in 1990.

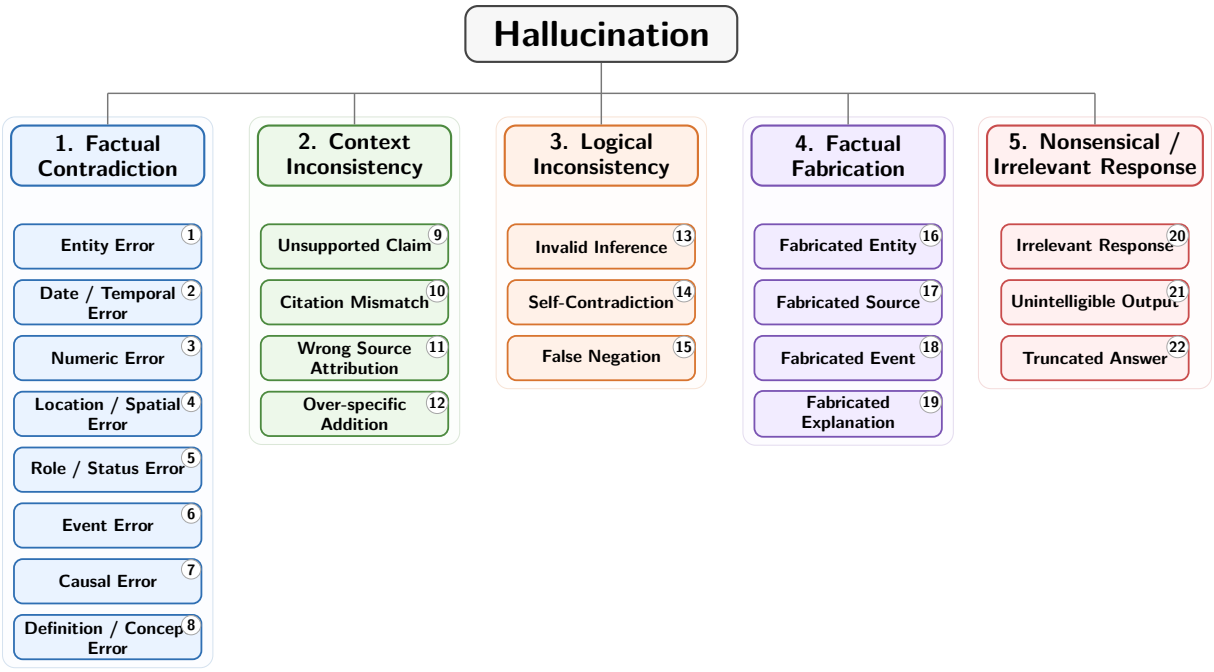


Figure 3: Hierarchical taxonomy of hallucination macro-types and observed micro-types.

3 Factual Contradiction – Numeric Error

Question

ما الطول التقريبي لنهر سانت لورانس، ومن أي بحيرة ينبع؟

What is the approximate length of the Saint Lawrence River, and from which lake does it originate?

Prediction

نهر سانت لورانس يمتد تقريبًا لـ 3,058 كيلومتر (1,900 ميل). وهو ينبع من بحيرة أونتاريو، التي هي واحدة من البحيرات الكبرى في أمريكا الشمالية.

The Saint Lawrence River extends for approximately 3,058 kilometers (1,900 miles). It originates from Lake Ontario, which is one of the Great Lakes of North America.

Gold answer

نحو 1,197 كم، وينبع من بحيرة أونتاريو.

Approximately 1,197 km, and it originates from Lake Ontario.

Explanation

الخطأ في الإجابة هو ذكر طول غير صحيح لنهر سانت لورانس. فطول النهر من بحيرة أونتاريو حتى المصب في خليج سانت لورانس نحو 1,197 كم، وليس 3,058 كم. وقد يشير الرقم 3,058 كم إلى منظومة البحيرات العظمى-سانت لورانس بأكملها، لا إلى النهر نفسه.

The error is the incorrect length given for the Saint Lawrence River. The river's length from Lake Ontario to its mouth in the Gulf of Saint Lawrence is approximately 1,197 km, not 3,058 km. The figure 3,058 km may refer to the entire Great Lakes–Saint Lawrence system, not to the river itself.

4 Factual Contradiction – Location / Spatial Error

Question

ما المدينة التي تسمى 'مدينة الموتى' في مصر؟

Which place is called the "City of the Dead" in Egypt?

Prediction

المدينة التي تُعرف باسم ``مدينة الموتى" في مصر هي مدينة الأقصر. ولكن، من الجدير بالذكر أن هذا الاسم ليس رسمياً وقد يكون مستخدماً بشكل غير رسمي بسبب الكثافة العالية للمواقع الأثرية والمدافن القديمة فيها.

The place known as the "City of the Dead" in Egypt is Luxor. However, it should be noted that this name is not official and may be used informally because of the high density of archaeological sites and ancient tombs there.

Gold answer

القرافة هي مدينة الموتى في قلب القاهرة الآن.

Al-Qarafa is the City of the Dead, located today in the heart of Cairo.

Explanation

The error is identifying Luxor as the "City of the Dead" in Egypt. This name refers to Al-Qarafa, the historic cemetery district in Cairo, not to the city of Luxor.

Macro-type	Micro-type	Definition
Factual Contradiction	Entity Error	The response gives an incorrect person, country, organization, object, or named entity instead of the verified answer.
	Date / Temporal Error	The response gives an incorrect date, year, period, duration, chronological order, or temporal relation.
	Numeric Error	The response gives an incorrect number, quantity, percentage, distance, length, count, or measurement.
	Location / Spatial Error	The response gives an incorrect place, region, country, spatial relation, or geographical location.
	Role / Status Error	The response assigns an incorrect role, title, status, function, affiliation, or relation to an entity.
	Event Error	The response describes a real or expected event incorrectly, assigns it to the wrong actor, time, place, or outcome, or confuses it with another event.
	Causal Error	The response gives an incorrect cause, effect, motivation, or explanatory relation between facts.
	Definition / Concept Error	The response confuses related concepts or gives a definition that contradicts the verified answer.
Context Inconsistency	Unsupported Claim	The response adds a claim that remains related to the question but is not supported by the verified evidence.
	Citation Mismatch	The response links a claim to the wrong citation, verse, hadith, book, source, or evidential reference.
	Wrong Source Attribution	The response attributes a statement to a source, scholar, text, or authority that does not support that statement.
	Over-specific Addition	The response adds details that are more specific than what the verified evidence allows, without necessarily inventing a new source or event.
Logical Inconsistency	Invalid Inference	The response uses related information but draws a conclusion that does not logically follow from it.
	Self-Contradiction	The response contains two or more claims that are mutually incompatible within the same answer.
	False Negation	The response incorrectly denies a true relation or states the opposite of the verified answer.
Factual	Fabricated Entity	The response introduces a non-attested entity and presents it as factual.
	Fabricated Source	The response invents a source, citation, reference, verse, hadith, book, or authority that does not exist or is not verified.
	Fabricated Event	The response describes an event that is not supported by the verified sources and appears to be invented.
	Fabricated Explanation	The response gives an explanation that sounds specific or authoritative but is not grounded in the verified evidence.
Nonsensical /	Irrelevant Response	The response is off-topic or does not address the question in a meaningful way.
	Unintelligible Output	The response is too unclear, incoherent, or broken to be interpreted as a valid factual answer.
	Truncated Answer	The response is incomplete or stops before providing a verifiable answer.

Table 6: Macro-types, micro-types, and definitions in the HALLUTRUTHQA hallucination taxonomy. The macro-level cues clarify the main distinction between wrong factual values, wrong grounding, wrong reasoning, invented factual content, and unusable responses.

5 Factual Contradiction – Role / Status Error

Question

من هو القائد الذي حرر الجزائر من الاستعمار الفرنسي سياسيًا؟

Who was the leader who politically liberated Algeria from French colonialism?

Prediction

القائد الذي حرر الجزائر من الاستعمار الفرنسي سياسيًا هو أحمد بن بلة. تولى رئاسة الحكومة المؤقتة للجمهورية الجزائرية في عام 1962 بعد مفاوضات إيفيان التي أدت إلى استقلال البلاد عن فرنسا.

The leader who politically liberated Algeria from French colonialism was Ahmed Ben Bella. He headed the Provisional Government of the Algerian Republic in 1962 after the Evian negotiations that led to Algeria's independence from France.

Gold answer

Benyoucef Benkhedda. يوسف بن خدة

Comment The model incorrectly attributes the presidency of the Provisional Government of the Algerian Republic during the Evian period to Ahmed Ben Bella. Ben Bella was imprisoned during the Evian negotiations, while Benyoucef Benkhedda headed the GPRA at that stage.

6 Factual Contradiction – Event Error

Question

ما المعركة التي حدثت بين علي بن أبي طالب وبين الخوارج؟

Which battle took place between Ali ibn Abi Talib and the Kharijites?

Prediction

معركة الجمل هي المعركة التي حدثت بين علي بن أبي طالب والخوارج. ولكن يجب التنويه إلى أن الخوارج لم يكونوا خصوم علي في هذه المعركة تحديداً، بل كانت معاوية بن أبي سفيان هو الخصم الرئيسي في معركة الجمل، بينما كانت معركة النهروان هي المعركة التي دارت بين علي بن أبي طالب والخوارج. لكن إذا كنت تقصد أي معركة بين علي والخوارج، فإن معركة النهروان هي الأكثر شهرة.

The Battle of the Camel was the battle that took place between Ali ibn Abi Talib and the Kharijites. However, it should be noted that the Kharijites were not Ali's opponents in this specific battle; rather, Muawiya ibn Abi Sufyan was the main opponent in the Battle of the Camel, whereas the Battle of Nahrawan was the battle that took place between Ali ibn Abi Talib and the Kharijites. But if you mean any battle between Ali and the Kharijites, then the Battle of Nahrawan is the most famous.

Gold answer

المعركة التي حدثت بين علي بن أبي طالب والخوارج هي معركة النهروان عام 38 هـ.

The battle that took place between Ali ibn Abi Talib and the Kharijites was the Battle of Nahrawan in 38 AH.

Explanation

The first error is identifying the Battle of the Camel as the battle between Ali ibn Abi Talib and the Kharijites, whereas the correct battle is the Battle of Nahrawan. The statement that Muawiya ibn Abi Sufyan was the main opponent in the Battle of the Camel is also incorrect; Muawiya is mainly associated with the Battle of Siffin, not the Battle of the Camel.

7 Factual Contradiction – Causal Error

Question

لماذا ينصح المرأة المرضع بعدم تناول الكحول؟

Why is a breastfeeding woman advised not to consume alcohol?

Prediction

يُسمح بالكحول أثناء الرضاعة بكميات معتدلة لأنه يُسرّع امتصاص الحديد في الحليب ولا يصل للرضيع.

Alcohol is allowed during breastfeeding in moderate amounts because it accelerates iron absorption in breast milk and does not reach the infant.

Gold answer

ينصح المرأة المرضع بعدم تناول الكحول لأن الكحول يمكن أن ينتقل إلى حليب الأم ويصل إلى الرضيع، وقد يؤثر في صحته ونموه ونمط نومه.

A breastfeeding woman is advised not to consume alcohol because alcohol can pass into breast milk and reach the infant, potentially affecting the infant's health, development, and sleep patterns.

Explanation

The error lies in giving a false causal justification for alcohol consumption during breastfeeding. Alcohol is not recommended because it does not accelerate iron absorption in breast milk, and the claim that it does not reach the infant is incorrect; alcohol can pass into breast milk and expose the infant.

8 Factual Contradiction – Definition / Concept Error

Question

ما هي الميتة الحلال والدم الحلال؟

What are the lawful dead animals and the lawful blood in Islamic law?

Prediction

الميتة الحرام هي ما مات بغير ذكاة شرعية، أما الميتة الحلال فهي ما ماتت ذكاة شرعية. أما الدم الحرام فهو الدم الذي لا يجوز أكله، مثل دم الحيوانات الميتة أو التي لم تذبح بشكل صحيح، بينما الدم الحلال هو دم الحيوانات التي ذبحت بطريقة شرعية. ولكن يجب التنويه إلى أن الإسلام يحرم تناول الدم بشكل عام، سواء كان من حيوان مذبوح بشكل صحيح أم لا.

Forbidden carrion is what dies without lawful slaughter, whereas lawful carrion is what dies through lawful slaughter. Forbidden blood is blood that may not be consumed, such as the blood of dead animals or animals not slaughtered properly, whereas lawful blood is the blood of animals slaughtered according to Islamic law. However, it should be noted that Islam generally forbids consuming blood, whether from a properly slaughtered animal or not.

Gold answer

الميتة الحلال هي السمك والجراد، والدم الحلال هو الكبد والطحال.

The lawful dead animals are fish and locusts, and the lawful blood refers to the liver and spleen.

Explanation

The error lies in giving an incorrect definition of lawful carrion and lawful blood. Lawful carrion does not mean an animal that died through lawful slaughter; rather, it refers to the recognized exceptions: fish and locusts. Similarly, lawful blood does not mean the blood of a properly slaughtered animal, but refers to the liver and spleen.

9 Context Inconsistency – Unsupported Claim

Question

ما هي أكبر جزيرة في العالم؟

What is the largest island in the world?

Prediction

جرينلاند هي الأكبر بمساحة تبلغ حوالي 2,166,086 كيلومتر مربع. ومع ذلك، يعتمد هذا على كيفية تعريف ``الجزر'' -- فجرينلاند تعتبر أحياناً شبه جزيرة بسبب اتصالها القاري مع كندا عبر مضيق دنمارك الصغير.

Greenland is the largest island, with an area of about 2,166,086 square kilometers. However, this depends on how "islands" are defined – Greenland is sometimes considered a peninsula because of its continental connection with Canada through the narrow Denmark Strait.

Gold answer

غرينلاند

Greenland.

Explanation

The model gives the correct main answer, but then adds an unsupported claim about Greenland being sometimes considered a peninsula because of a supposed continental connection with Canada. This extra claim is not supported by the question or the reference answer and introduces unnecessary, ungrounded information.

10 Factual Contradiction – Location / Spatial Error

Question

ما اسم البحيرة الواقعة على حدود زيمبابوي وزامبيا والمعروفة بأنها من أكبر البحيرات الاصطناعية؟

What is the name of the lake located on the border between Zimbabwe and Zambia, known as one of the largest artificial lakes?

Prediction

البحيرة التي تشير إلى هي بحيرة كاريبا، وهي تقع على الحدود بين زيمبابوي وزامبيا، وتشتهر بوجود شلالات فيكتوريا الشهيرة.

The lake referred to is Lake Kariba, located on the border between Zimbabwe and Zambia, and it is famous for the presence of Victoria Falls.

Gold answer

الاسم هو بحيرة كاريبا (Lake Kariba)

Error. The model identifies the correct lake, but adds a false geographical association with Victoria Falls.

11 Factual Contradiction – Citation Mismatch

Question

هل سنّ النبي صلى الله عليه وسلم الإسراع بتجهيز الميت أم الإبطاء به؟

Did the Prophet, peace be upon him, prescribe hastening the preparation of the deceased or delaying it?

Prediction

لم يسن النبي صلى الله عليه وسلم الإسراع أو الإبطاء بشكل خاص بتجهيز الميت، ولكن يُفضل الإسراع في تجهيزه دون تأخير طويل، وذلك لما روي عن عائشة رضي الله عنها أنها قالت: كان رسول الله صلى الله عليه وسلم إذا حضرته الوفاة جعل عنده شيئاً من التراب فيمسحه بوجهه ويقول: بِسْمِ اللَّهِ وَبِعِزَّةِ اللَّهِ، وَأَعُوذُ بِاللَّهِ .

The Prophet, peace be upon him, did not specifically prescribe either hastening or delaying the preparation of the deceased; however, it is preferable to prepare the deceased quickly without long delay, based on a report attributed to Aisha, may Allah be pleased with her, stating that when the Prophet was near death, he would place some dust near him, wipe his face with it, and say: In the name of Allah and by the might of Allah, and I seek refuge in Allah from....

Gold answer

سنّ النبي صلى الله عليه وسلم الإسراع بتجهيز الميت ودفنه، وعدم تأخيره بلا حاجة.

The Prophet, peace be upon him, prescribed hastening the preparation and burial of the deceased, and not delaying it without need.

Explanation

الخطأ الأول هو نفي وجود سنة خاصة بالإسراع في تجهيز الميت، بينما السنة هي الإسراع بتجهيزه ودفنه وعدم تأخيره بلا حاجة. أما الخطأ الثاني فهو الاستدلال برواية غير مناسبة لا تدعم الحكم المتعلق بتجهيز الميت.

The first error is denying that there is a specific prophetic practice of hastening the preparation of the deceased, whereas the Sunnah is to prepare and bury the deceased promptly without unnecessary delay. The second error is the use of an inappropriate and incomplete narration that does not support the ruling about preparing the deceased.

12 Context Inconsistency – Wrong Source Attribution

Question

أشار القرآن الكريم إلى عدة مهن كانت تزاول في الأمم السابقة، منها مهنة بناء السفن، فما الآيات التي أشارت إلى هذه المهنة؟ ومن هو النبي الذي زاول هذه المهنة؟

Prediction

الآية 14 من سورة هود، موسى عليه السلام كان يزاول مهنة بناء السفن؛ الآية 41 من سورة الأنبياء، الآية 40 من سورة هود؛ النبي الذي زاول مهنة بناء السفن هو موسى عليه السلام

Gold answer

هود: 37، 38؛ نوح عليه السلام

Error. The answer provides wrong Qur'anic references and assigns the action to the wrong prophet.

13 Logical Inconsistency – Invalid Inference

Question

ما السرية التي أرسل فيها سبعون من الصحابة فقُتل أكثرهم عند بئر معونة؟

Prediction

سرية الخرار، بقيادة عبد الله بن حرام، أكثر من سبعين

Gold answer

سرية بئر معونة، وتسمى أيضاً سرية القراء.

Error. The answer gives unsupported details and an incorrect inference.

14 Factual Fabrication – Fabricated Entity

Question

ما اسم الحوض الذي يضم أعمق نقطة في بحر بافين شمال كندا، وما عمقه التقريبي؟

Prediction

حوض نيلسون، وعمقه التقريبي حوالي 1,450 متر

Gold answer

حوض بافين بعمق نحو 2,400 م

Error. The model introduces an incorrect or fabricated basin name.

15 Factual Fabrication – Fabricated Source / Quote

Question

ما البلد الذي ذُكر اسمه في القرآن الكريم؟

Which country is mentioned by name in the Qur'an?

Prediction

بلد واحد فقط ذُكر اسمه في القرآن الكريم وهو مصر، وذلك في سورة يوسف الآية 47: وَقَالَ الْمَلِكُ ائْتُونِي بِهِ فَلَمَّا جَاءَهُ الرَّسُولُ قَالَ ارْجِعْ إِلَىٰ رَبِّكَ فَسْأَلْهُ مَا بَالُ النِّسْوَةِ اللَّاتِي قَطَّعْنَ أَيْدِيَهُنَّ إِنَّ رَبِّي بِكَيْدِهِنَّ عَلِيمٌ

Only one country is mentioned by name in the Qur'an, namely Egypt, in Surah Yusuf, verse 47: "And the king said: Bring him to me. Then, when the messenger came to him, he said: Return to your lord and ask him about what belongs to the two breasts..."

Gold answer

مصر

Egypt.

Error. The model gives the correct short answer, but fabricates a Qur'anic-style quotation and falsely attributes it to Surah Yusuf, verse 47.

16 Nonsensical Response – Irrelevant / Broken Output

Question

من هو الملك الهاشمي الذي يعرف بلقب الملك المؤسس؟

Prediction

العرابيين

Gold answer

الملك عبدالله الأول

Error. The output is malformed and does not answer the question.

17 Factual Fabrication – Date / Temporal Error

Question

ما المدينة التي تسمى ``عاصمة الثقافة العربية 2024"؟

Prediction

مدينة سطيف الجزائرية

Gold answer

المدينة التي حملت لقب `عاصمة الثقافة العربية لعام 2024" هي مدينة طرابلس اللبنانية. كانت مقررة أصلاً لعام 2023، لكنها تأجّلت بسبب الأزمة الاقتصادية اللبنانية. أما مدينة سطيف الجزائرية فقد حملت لقب `عاصمة الثقافة العربية" في عام 2007.

Error. The model fabricates the selected city by answering Setif, Algeria, whereas the correct city for 2024 is Tripoli, Lebanon. Setif was previously designated as the Arab Capital of Culture in 2007.

F Explanation Evaluation Prompt

Prompt for Explanation Evaluation

We evaluate the quality of the model explanation for hallucinated answers. This prompt is used only when both the gold label and the predicted label are *hallucination*.

The evaluation is based on two dimensions:

1. Error Identification (0–1)
   The explanation should identify the hallucinated span or claim and clearly explain why it is incorrect.

2. Factual Correction (0–1)
   The explanation should provide the correct factual information or a valid correction of the hallucinated content.

Each dimension is scored independently on a continuous scale from 0 to 1, where 0 means completely incorrect or missing, and **1** means fully correct and complete. The scoring should be consistent across examples: explanations with similar quality should receive similar scores.

The evaluator must return JSON only:

```
{
  "error_identification": float,
  "factual_correction": float,
  "reason": "short justification"
}
```